

CBMR SKI PATROL WEB APP

Joshua Clyncke, Rhys Jordan & Nate Rengo

Crested Butte Mountain Resort Ski Patrol

CLIENT

Krista Hildebrandt

- CBMR Ski patroller for 35 years
- Snow Safety Director
- Western Alumni



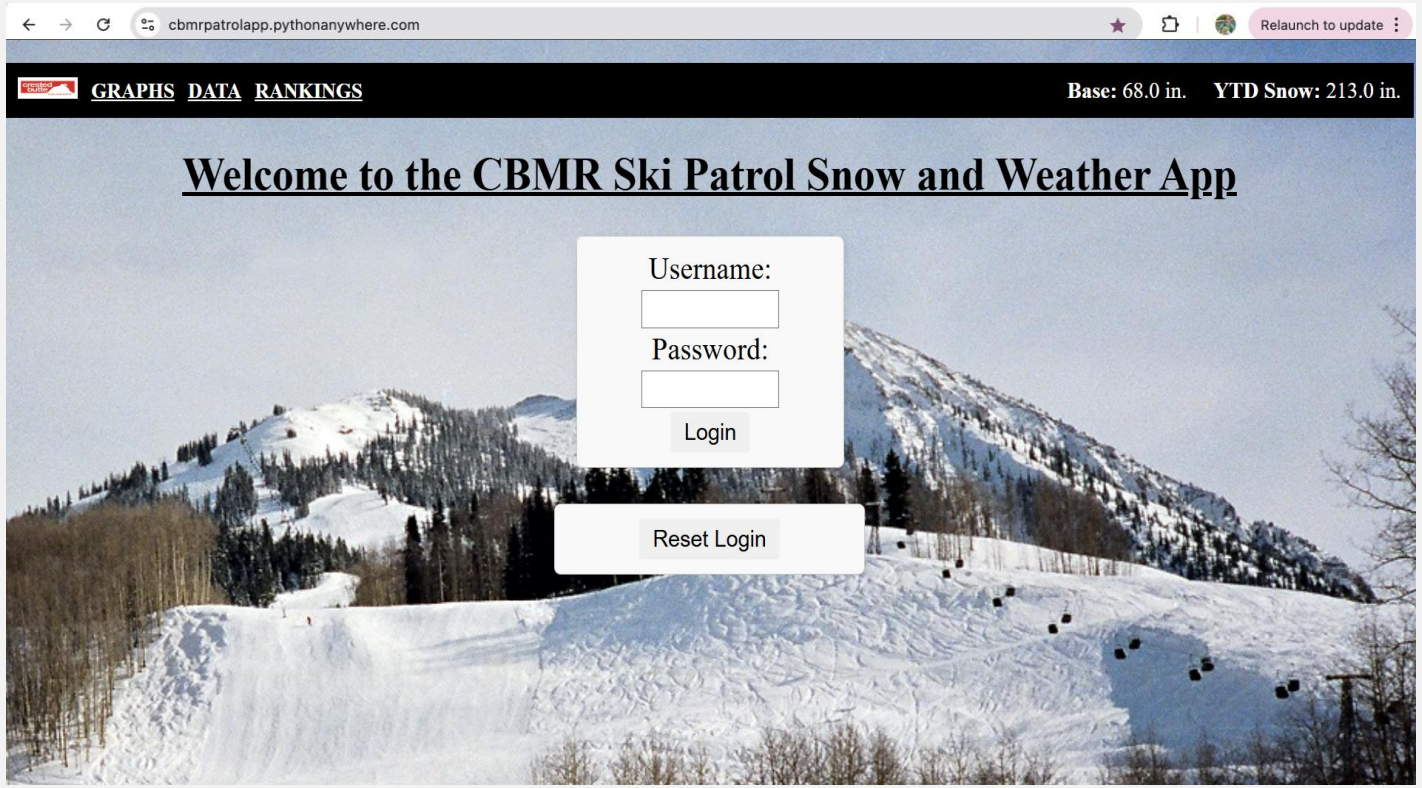
CBMR SKI PATROL

- Ski Patrol leads
  - Avalanche mitigation
  - Medical and safety incident responses
  - Trail demarcation
  - Condition monitoring efforts



INTRO

Produce an online application for recording weather and snow data for CBMR Ski Patrol.



WHY DOES SKI PATROL TRACK WEATHER?

- **Weather Conditions**
  - What will the day look like? Temperatures, Snowfall, Sunlight
- **Avalanche Problems**
  - How dangerous is the snowpack? Can we open terrain?
- **Opening and Closing Terrain**
  - When can we open terrain? When was it open last?
- **Skier Safety**
  - Does weather pose a threat to skiers? The lifts?
- **Historical/Seasonal Records**
  - What day has the greatest 24hr snow fall? What was the best season?



RESEARCH







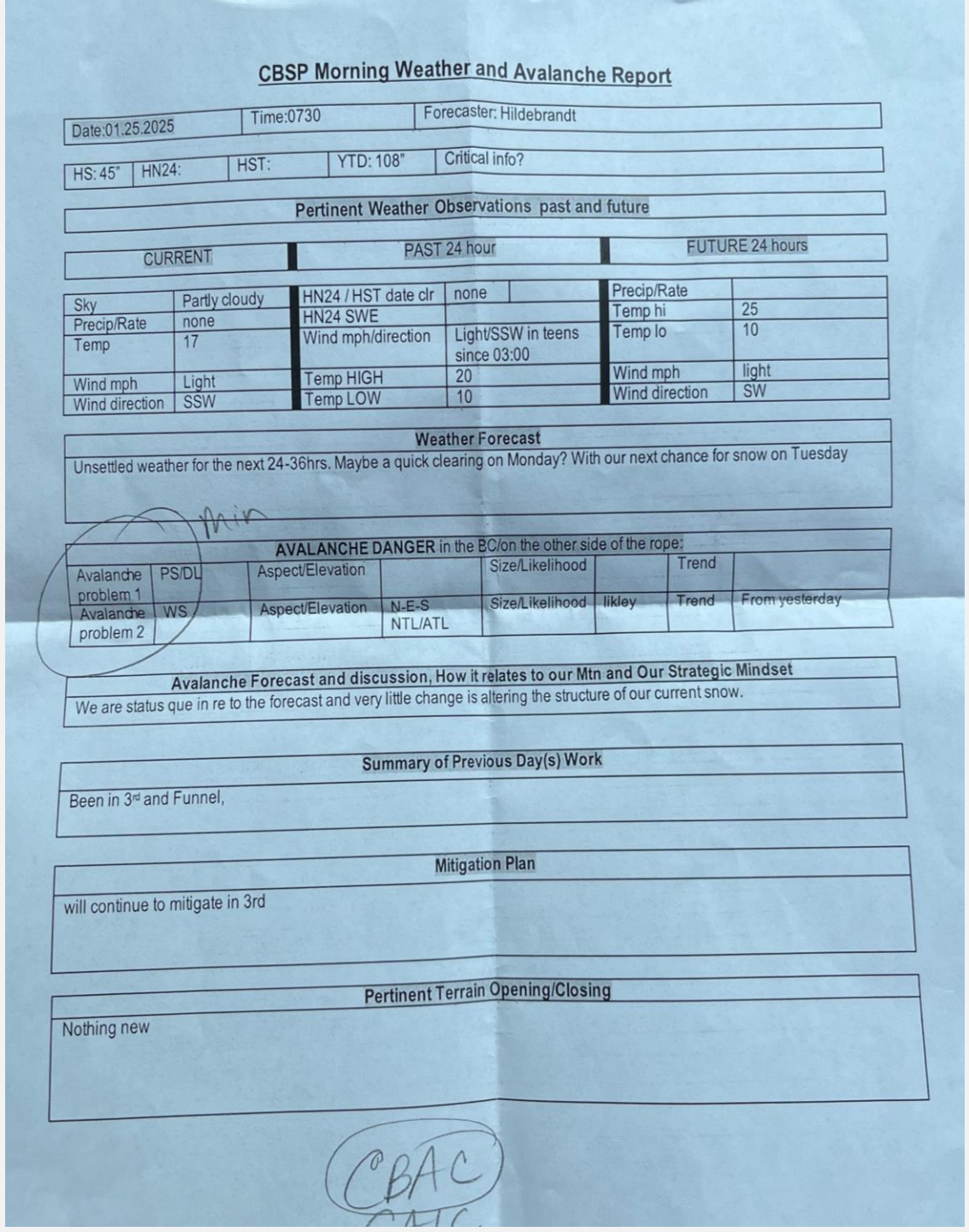


DAILY WORKFLOW

1. Record data in the field
2. Brief Ski Patrol
3. Patrol open terrain and mitigate closed terrain
4. Debrief Patrol and Brief Mountain Ops
5. Plan for next day

AM Form

PM Form



CBSP Morning Weather and Avalanche Report

Date:01.25.2025	Time:0730	Forecaster: Hildebrandt

HS: 45"	HN24:	HST:	YTD: 108"	Critical info?

Pertinent Weather Observations past and future

CURRENT		PAST 24 hour		FUTURE 24 hours	
Sky	Partly cloudy	HN24 / HST date clr	none	Precip/Rate	
Precip/Rate	none	HN24 SWE		Temp hi	25
Temp	17	Wind mph/direction	Light/SSW in teens since 03:00	Temp lo	10
Wind mph	Light	Temp HIGH	20	Wind mph	light
Wind direction	SSW	Temp LOW	10	Wind direction	SW

Weather Forecast

Unsettled weather for the next 24-36hrs. Maybe a quick clearing on Monday? With our next chance for snow on Tuesday

AVALANCHE DANGER in the BC/on the other side of the rope:

Avalanche problem 1	PS/DL	Aspect/Elevation		Size/Likelihood		Trend	
Avalanche problem 2	WS	Aspect/Elevation	N-E-S NTL/ATL	Size/Likelihood	likley	Trend	From yesterday

Avalanche Forecast and discussion, How it relates to our Mtn and Our Strategic Mindset

We are status que in re to the forecast and very little change is altering the structure of our current snow.

Summary of Previous Day(s) Work

Been in 3rd and Funnel,

Mitigation Plan

will continue to mitigate in 3rd

Pertinent Terrain Opening/Closing

Nothing new

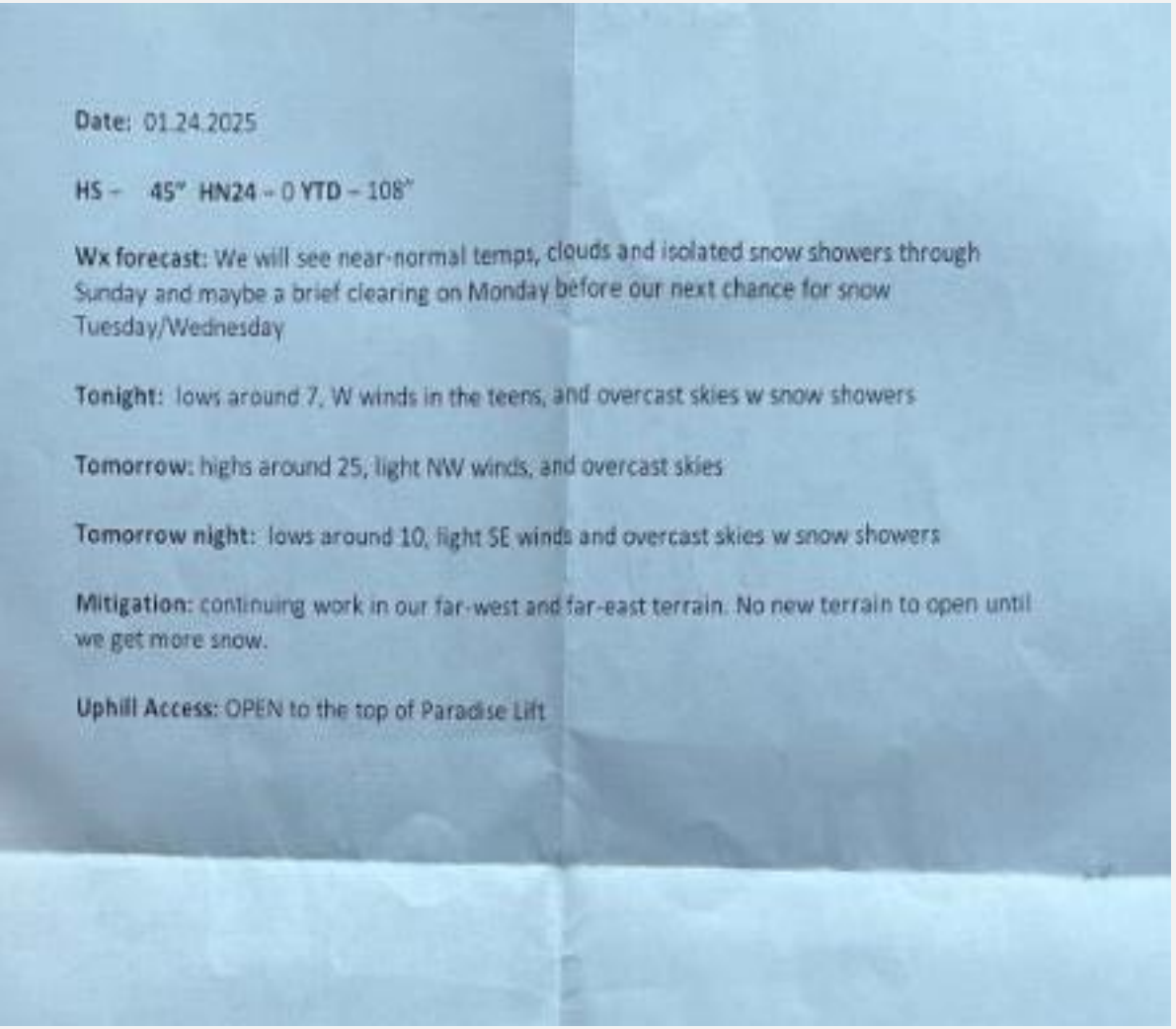
CBAC

AM FORM

- Tracks snow totals
- Tracks and predicts weather changes
  - Current, Past 24hr, Future 24hr
  - Weather Forecast
- Tracks avalanche dangers
- Concisely presents weather data
- Summarizes previous days work
- Summarizes mitigation plan
- Shared at the morning briefing

AM FORM WORKFLOW

1. Gather data from online and weather stations
2. Input into an AM Form in Word Doc
3. Enter same info into Excel Sheet
4. Export Word Doc to a PDF
5. Email and Print the AM Form
6. Save AM PDF in Google Drive



Date: 01.24.2025

HS – 45" HN24 – 0 YTD – 108"

Wx forecast: We will see near-normal temps, clouds and isolated snow showers through Sunday and maybe a brief clearing on Monday before our next chance for snow Tuesday/Wednesday

Tonight: lows around 7, W winds in the teens, and overcast skies w snow showers

Tomorrow: highs around 25, light NW winds, and overcast skies

Tomorrow night: lows around 10, light SE winds and overcast skies w snow showers

Mitigation: continuing work in our far-west and far-east terrain. No new terrain to open until we get more snow.

Uphill Access: OPEN to the top of Paradise Lift

PM FORM

- Summarize events over the day
- Snow Totals
- Weather forecast
- Plans for tomorrow
- Shared with mountain and lift operations

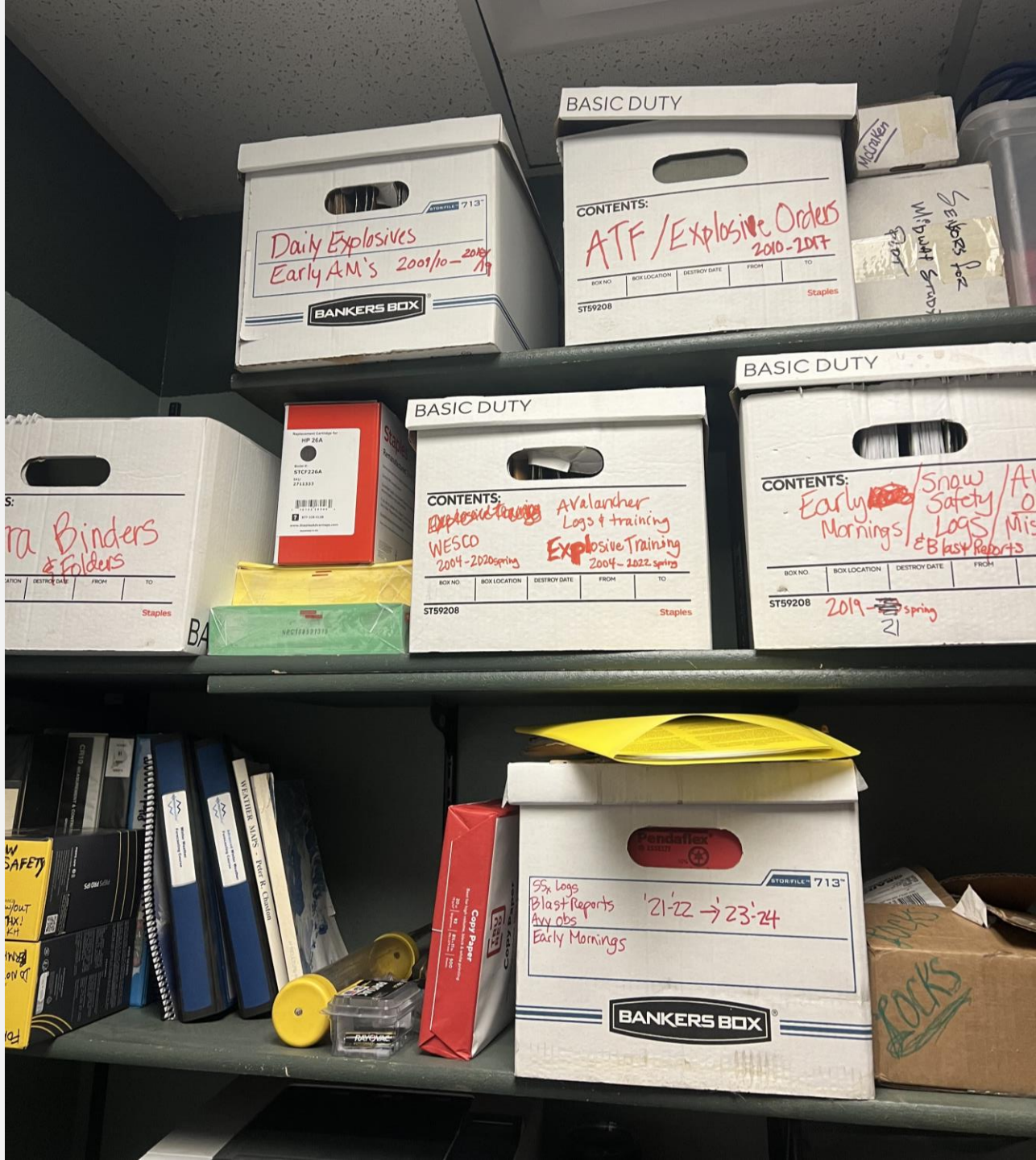
PM FORM WORKFLOW

1. Input data into an PM Form in Word Doc
2. Manually copy data from AM Form
3. Export Word Doc to a PDF
4. Email PM Form
5. Enter same information into InfoEx

WHAT DAY HAD THE GREATEST 24HR SNOWFALL?

- Currently Data storage exists in two places:
  - Boxes of paper forms up to 2000
  - An Excel document from 2000-Now

- No consistency in the data
- No way to use the data



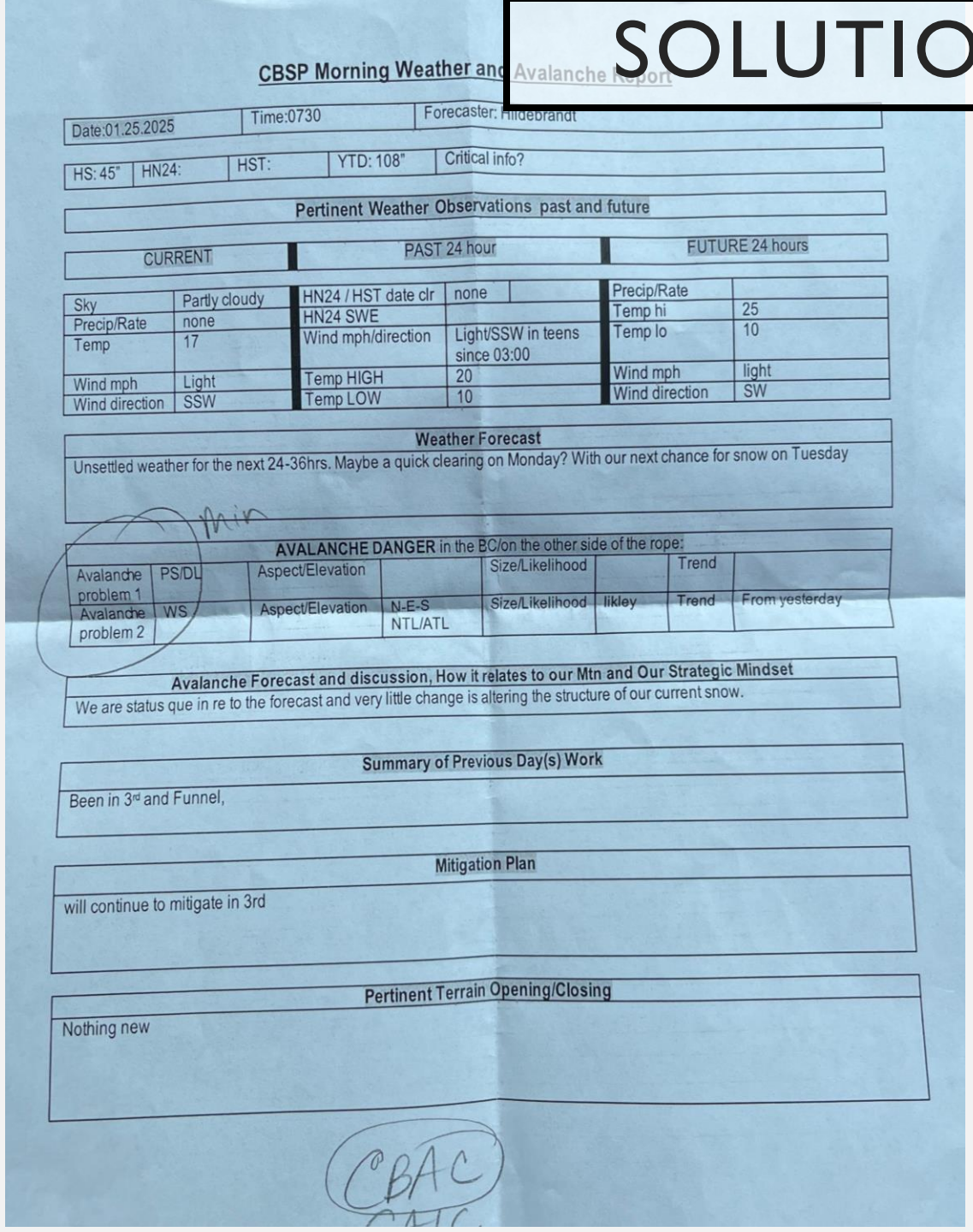
REQUIREMENTS

- Fillable AM and PM forms with respective data
- Ability to generate pdfs of both AM and PM forms
- Visualize important trends in data
- Compile relevant statistics
- Accessible on a computer
- Store data for long term usage

SOLUTION OVERVIEW

- **Goal:** To reduce the time and energy spent by CBMR entering their weather data
- Create a website
  - To enter, record and create pdfs of their weather data
  - Produces visuals and summaries of existing data

SOLUTION - AM FORM



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Mitigation Plan

will continue to mitigate in 3rd

Pertinent Terrain Opening/Closing

Nothing new

CBAC

CBSP Morning Weather and Avalanche Report

Date: 04/05/2025 Time: 5:48 AM Forecaster: Hildebrandt

HS: 70.0	HN24: 1.0	SWE: 0.1	HST: None	YTD Snow: 213.0	YTD SWE: 18.29
Critical Info?					

Pertinent Weather Observations Past and Future

Current		PAST 24 hour				FUTURE 24 hours	
Sky	few	HST		HN24	1.0	Precip/Rate	NO
Precip/Rate	NO	Date Cleared		HN24 SWE	0.1	Temp HIGH	30.0
Temp	12.0	Settled		HN24 %	10.0	Temp LOW	20.0
Wind mph	light	Wind mph	light	Temp HIGH	22.0	Wind mph	light
Wind Direction	NE	Wind Direction	NE	Temp LOW	10.0	Wind Direction	NNE
Gust MPH	5.0	Gust MPH		13.0		Gust MPH	20.0
Gust Direction	NE	Gust Direction		NE		Gust Direction	NE
Gust Time	2:15 AM	Gust Time		2:15 AM		Gust Time	12:00 AM

Notes: Winds have been light and variable from The northerly directions and snow ended around midnight last night

Weather Forecast

As the unsettled weather pattern of the last week departs our state, we will see possible snow showers with clearing skies and warming temps. Each day looks a little warmer with Friday being another possible record warm day. Tomorrow looks to be an excellent closing day with partly cloudy skies and temps in the 30s

Resort Danger	low	Backcountry Danger	moderate	PWL	Type:	Date:

Avalanche Problems

	Location	Problem	Elevation: Aspect	Size	Likelihood
Problem 1	Resort	Wind Slab	ATL: NE-E-SE NTL: NE-E-SE BTL: NE-E-SE	1	unlikely

Avalanche Forecast and discussion, How it relates to our Mtn and Our Strategic Mindset

We did see very small wind slabs in a few very specific locations on our mountain but they were so small they did not require any mitigation. The snow that fell the last few days improved skiing so much that yesterday our traverses may have been the best of the season. After a bit of mitigation work yesterday we were able to get Many Names reopened. We will be in maintenance mode for the last 2 days of our season and may even remove a few of the interior ropes we deployed during the warm up several weeks ago. We are status que and no mitigation work is needed to maintain our current boundaries.

Summary of Previous Day(s) Work

We did mitigation work in Teo2 and reopened Many Names. We enjoyed the day by skiing the softer snow and maybe ran 2 10-50s.

Mitigation Plan

Today we will do a quick westside gun mission before re-opening the westside (all gates were closed on sweep). No other mitigation work is needed but we may pull interior east river ropes and the few spur ropes in the Tbar zones as they are no longer needed.

Pertinent Terrain Opening/Closing

Westside and Peak will start closed until westside gun mission is complete

SOLUTION - AM FORM

- Create AM form which they can fill out with all the needed information
- The AM form will auto-populate with necessary information from the day before
- When an AM form in filled out they can generate a pdf version
- The information is stored for future use

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Wind mph	light	Wind mph	light	Temp HIGH	22.0	Wind mph	light
Wind Direction	NE	Wind Direction	NE	Temp LOW	10.0	Wind Direction	NNE
Gust MPH	5.0	Gust MPH		13.0		Gust MPH	20.0
Gust Direction	NE	Gust Direction		NE		Gust Direction	NE
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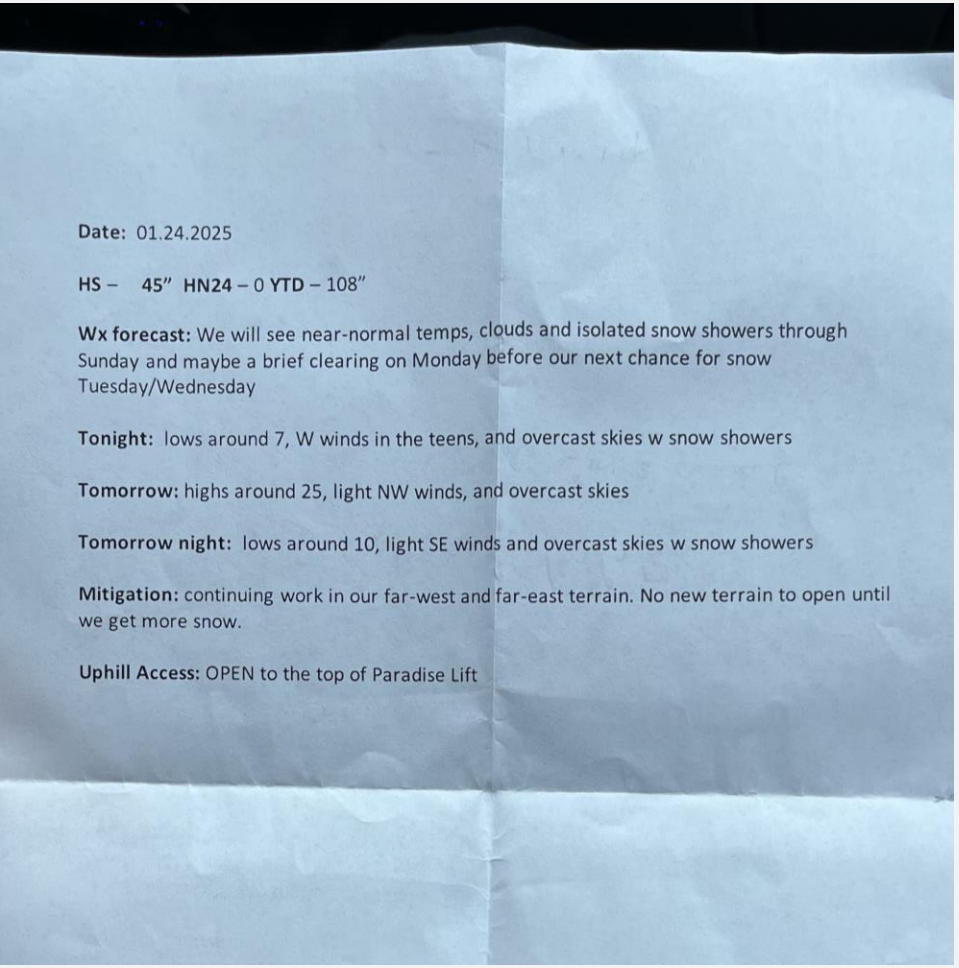
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Uphill Access: OPEN to the top of Paradise Lift

CBSP Evening Report

Date: 03/28/2025 Time: 2:13 PM Forecaster: nate

HS: 62.0	HN24: 0.0	YTD Snow: 197.0	YTD SWE: 16.89	Uphill Access: Open to top of Paradise

Weather Forecast
Clouds

Tonight, Tomorrow, Tomorrow night
tonight tomorrow etc

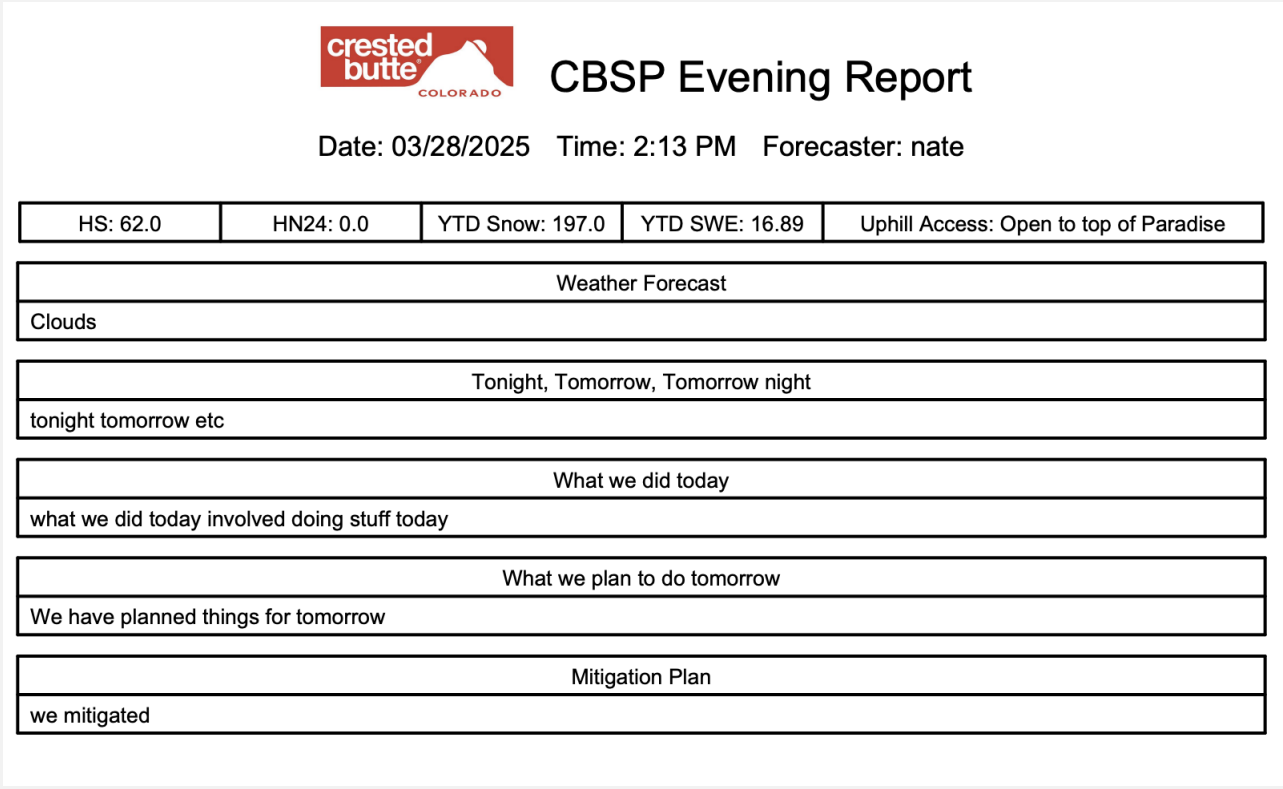
What we did today
what we did today involved doing stuff today

What we plan to do tomorrow
We have planned things for tomorrow

Mitigation Plan
we mitigated

SOLUTION - PM FORM

- Create PM form which they can fill out with all the needed information
- The PM form will auto-populate with necessary information from earlier in the day
- When an PM form in filled out they can generate a pdf version



crested butte COLORADO

CBSP Evening Report

Date: 03/28/2025 Time: 2:13 PM Forecaster: nate

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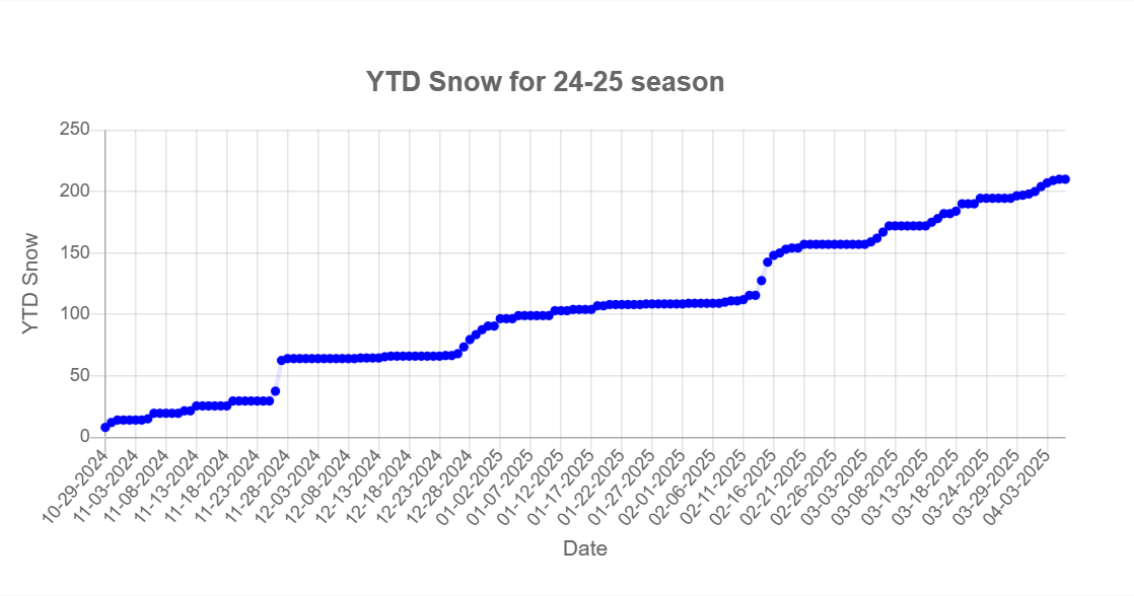
Weather Forecast
Clouds

Tonight, Tomorrow, Tomorrow night
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What we did today
what we did today involved doing stuff today

What we plan to do tomorrow
We have planned things for tomorrow

Mitigation Plan
we mitigated



Metric	Measure	Date
Greatest 24hr Snowfall	25.0 in.	2024-11-27
Coldest Temperature	-15.0°	2025-01-21
Deepest Snowpack	77.0 in.	2024-03-30

SOLUTION - WHAT DAY HAD THE GREATEST 24HR SNOWFALL?

- **Great 24hr Snowfall:**
  - **25 inches on November 27, 2024**
- Produce useful graphs
  - Show snowfall trends across a season
  - Show temperature trends
- Produce other useful statistics
  - Coldest Day:
    - -15° on January 12, 2025
  - Deepest Snowpack:
    - 77 inches on March 30, 2024

FRONT END



BACK END



Flask



SQL





HOSTING

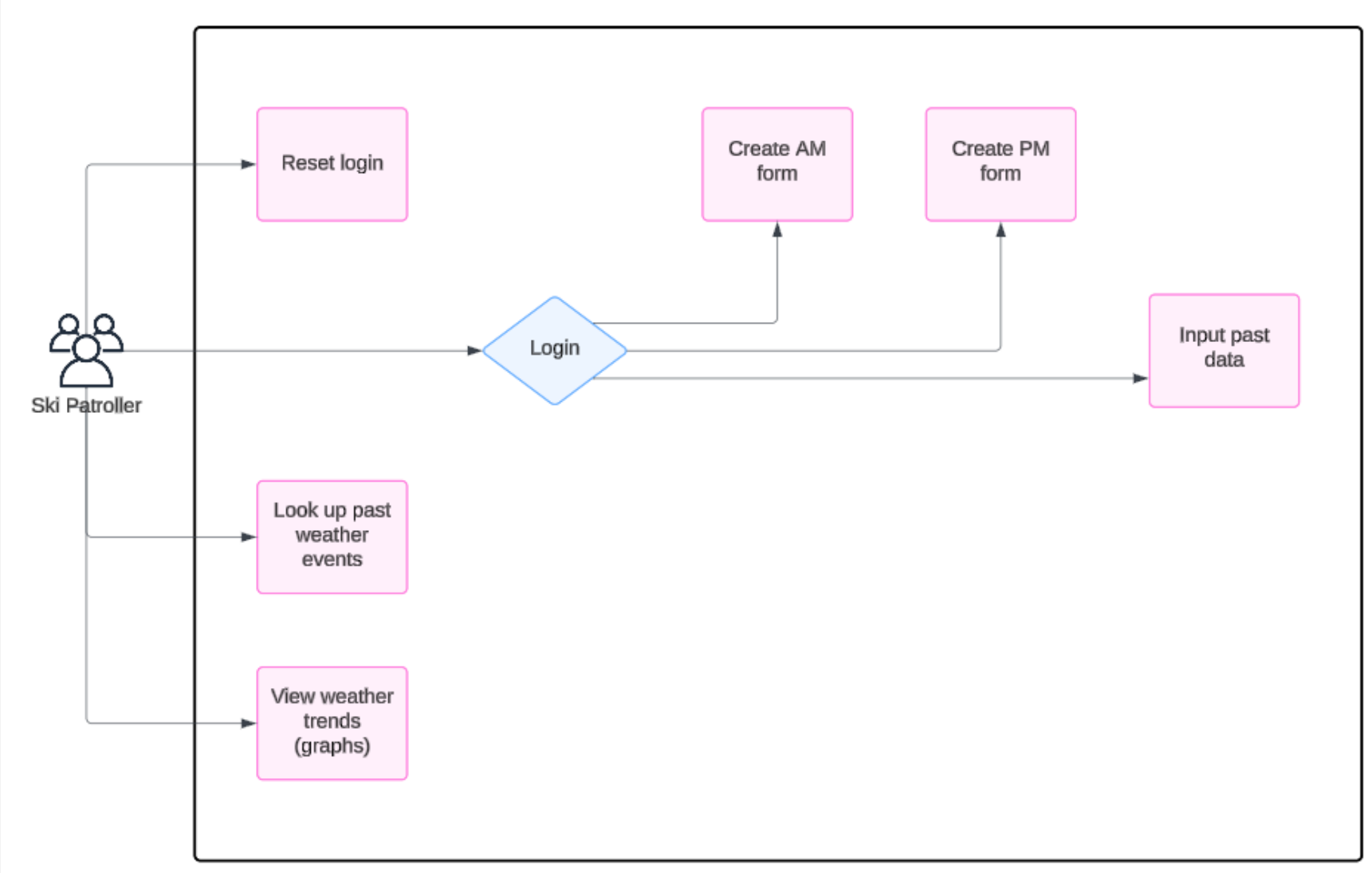




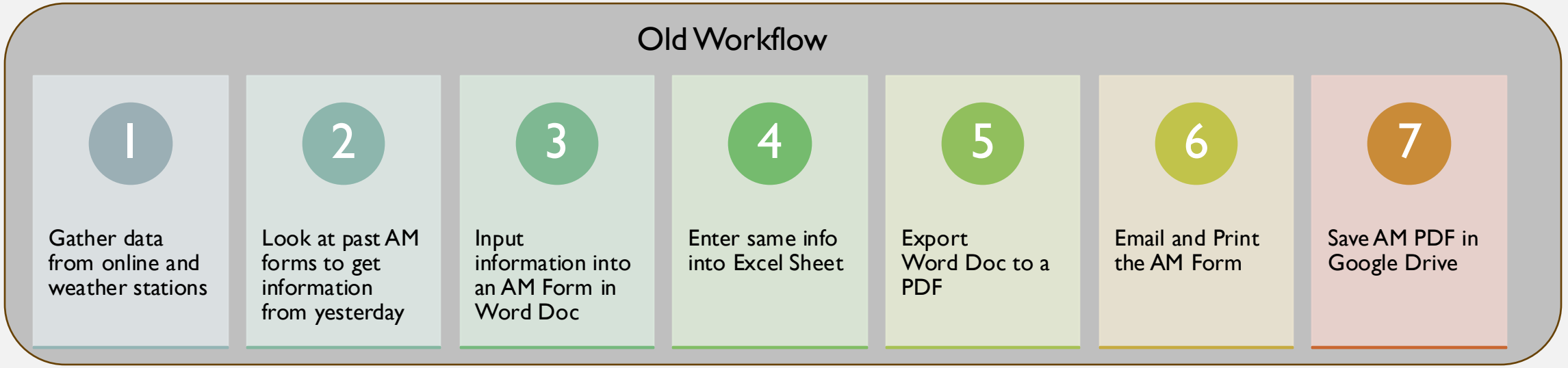




USE CASE DIAGRAM



AM WEBSITE WORKFLOW



AM WEBSITE WORKFLOW

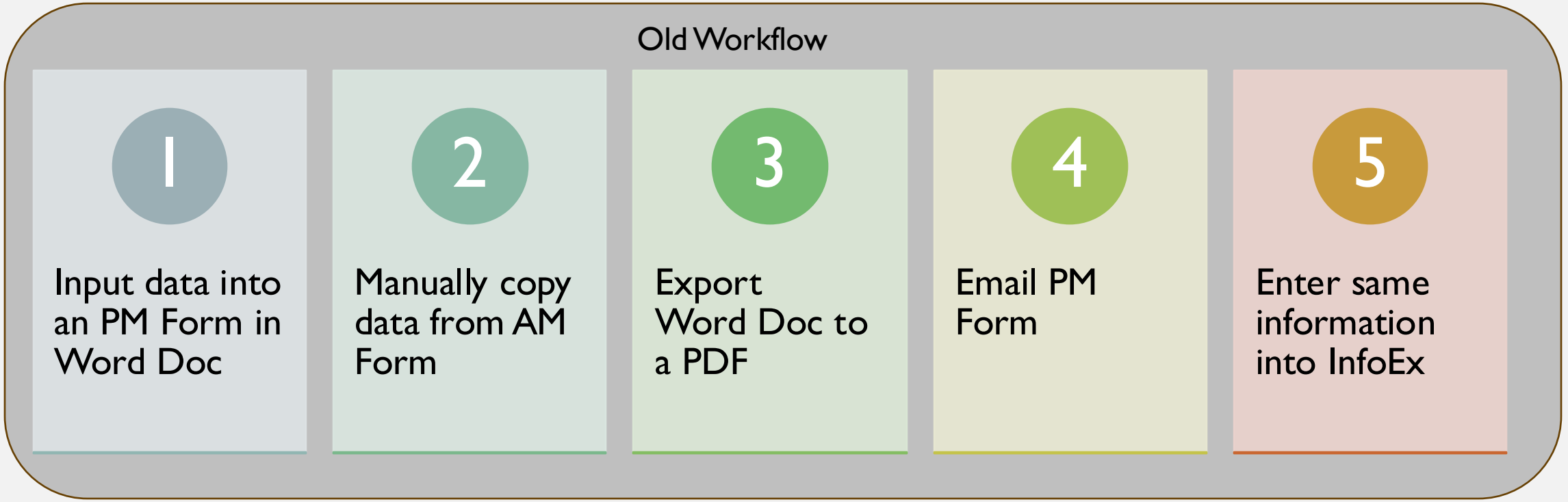
Old Workflow

1. Gather data from online and weather stations
2. ~~Look at past AM forms to get information from yesterday~~
3. Input information into an AM Form in Word Doc
4. ~~Enter same info into Excel Sheet~~
5. ~~Export Word Doc to a PDF~~
6. Email and Print the AM Form
7. ~~Save AM PDF in Google Drive~~

New Workflow

1. Gather data from online and weather stations
2. Database populates the AM form with relevant information
3. Input and update information in AM form
4. Submit AM form
   - Populates Database
   - Generates pdf
5. Email and Print the AM Form

PM WEBSITE WORKFLOW



PM WEBSITE WORKFLOW

Old Workflow

1. Input data into an PM Form in Word Doc
2. ~~Manually copy data from AM Form~~
3. ~~Export Word Doc to a PDF~~
4. Email PM Form
5. ~~Enter same information into InfoEx~~

New Workflow

1. Input data into an PM Form
2. Database populates the PM form with relevant information from the AM form
3. Submit PM form
   - Populates Database
   - Generates pdf
4. Email PM Form

LIVE DEMO

https://cbmrpatrolapp.pythonanywhere.com/

CHALLENGES AND RESOLUTIONS: HOSTING

- Working with pythonanywhere:
  - Server time
  - Database management

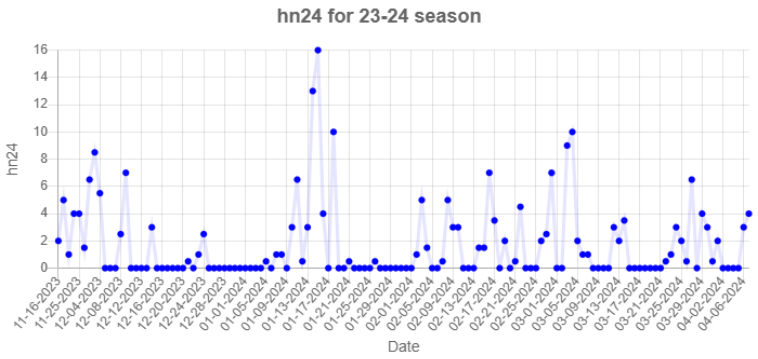


- Resolution:
  - Trial and error/ Getting used to pythonanywhere
  - Refactoring

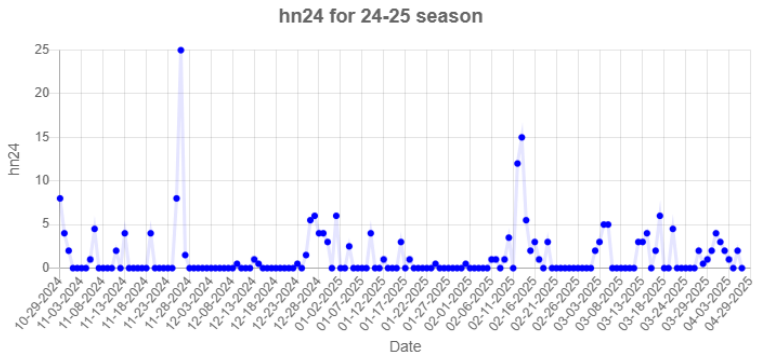
CHALLENGES AND RESOLUTIONS: GRAPHING

- Lining up the dates when trying to graph data:
  - 23-24 season started on 11/16/2023
  - 24-25 season started on 10/29/2024
  - To accurately graph data for both season on the same graph would have to match up dates
- Resolution:

Graph 1:

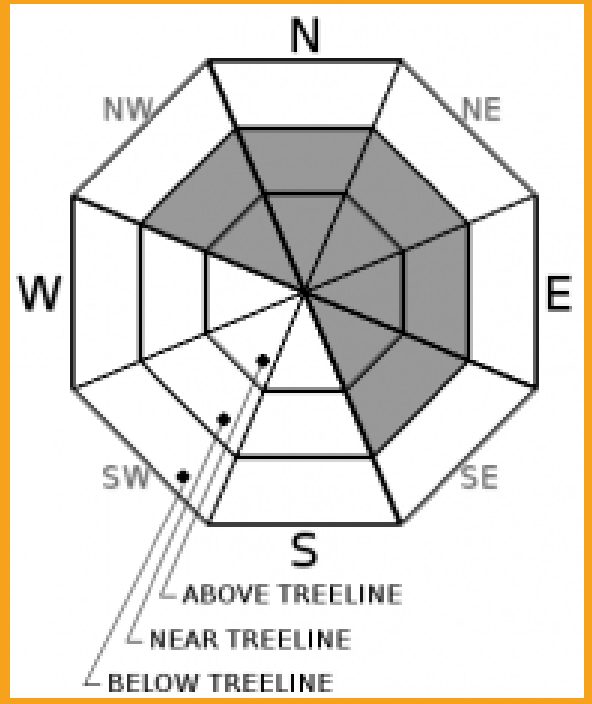


Graph 2:



CHALLENGES AND RESOLUTIONS: AVALANCHE ROSE

Industry



First Design

AVALANCHE DANGER in the BC/on other side of the rope

Danger Rating Resort Danger Rating Backcountry
Location Problem 1 Aspect Elevation Size Likelihood

Add Second Problem

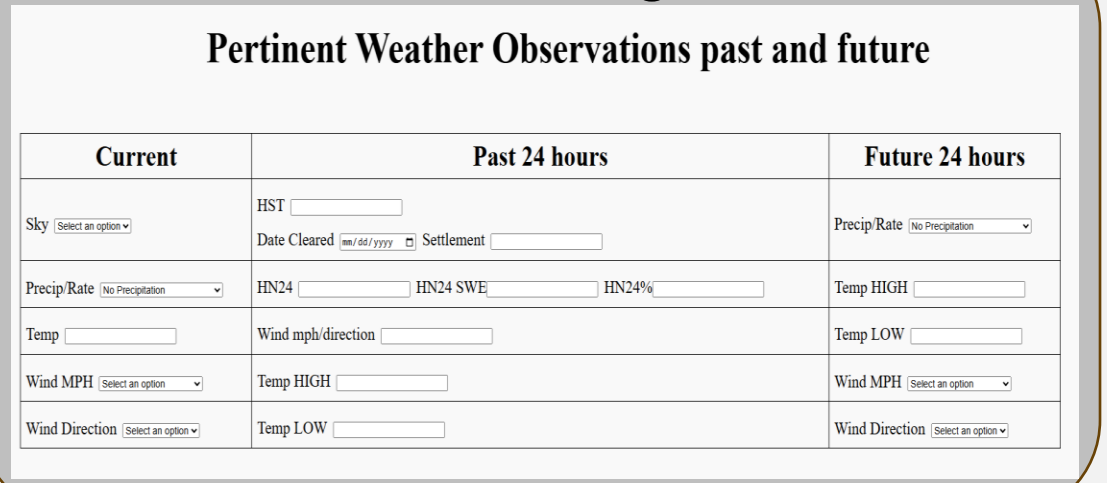
Final Design

Elevation / Aspect	N (0°)	NE (45°)	E (90°)	SE (135°)	S (180°)	SW (225°)	W (270°)	NW (315°)
Below Tree Line:	☐	☐	☐	☐	☐	☐	☐	☐
Near Tree Line:	☐	☐	☐	☐	☐	☐	☐	☐
Above Tree Line:	☐	☐	☐	☐	☐	☐	☐	☐

CLIENT DEMO FEEDBACK

- Input fields track data Client wants
- Add fields they weren't already tracking
  - Year to Date Snow Water Equivalent (YTD SWE)
  - Peak Wind Gusts (Current, Past, Future)
  - Persistent Weak Layer

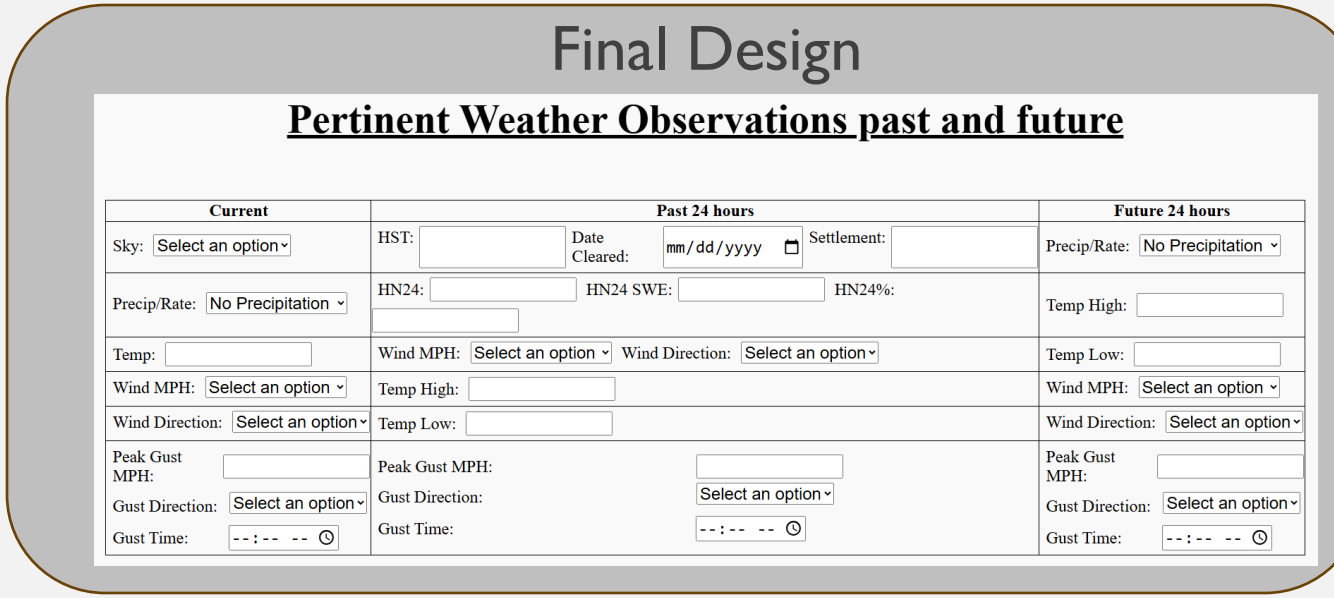
First Design



Pertinent Weather Observations past and future

Current	Past 24 hours	Future 24 hours
Sky Select an option	HST Date Cleared mm/dd/yyyy Settlement	Precip/Rate No Precipitation
Precip/Rate No Precipitation	HN24 HN24 SWE HN24%	Temp HIGH
Temp	Wind mph/direction	Temp LOW
Wind MPH Select an option	Temp HIGH	Wind MPH Select an option
Wind Direction Select an option	Temp LOW	Wind Direction Select an option

Final Design

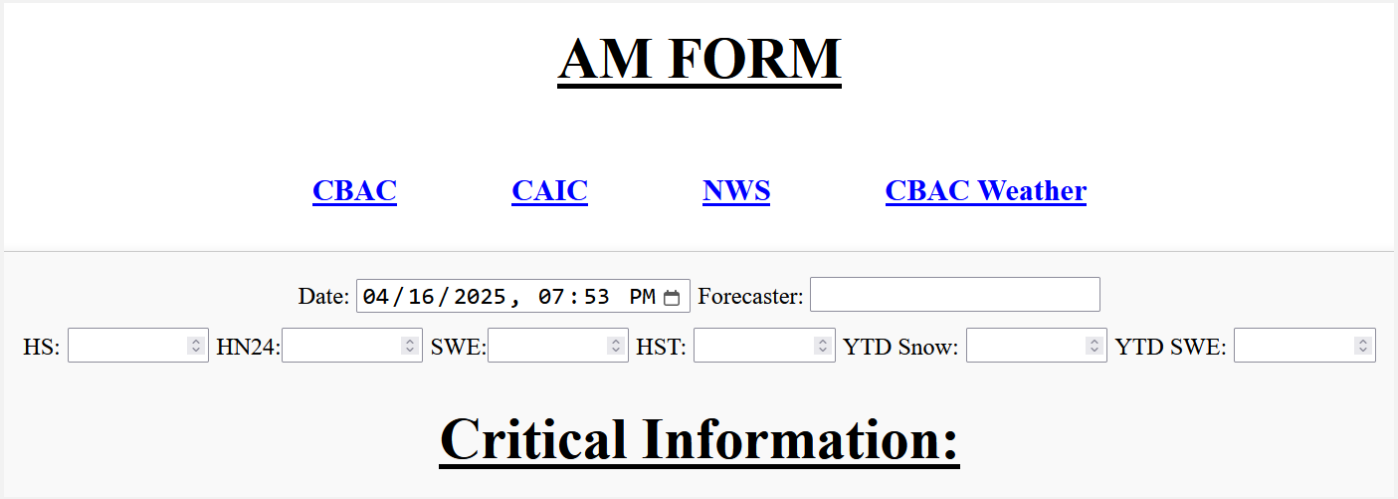


Pertinent Weather Observations past and future

Current	Past 24 hours	Future 24 hours
Sky: Select an option	HST: Date Cleared: mm/dd/yyyy Settlement:	Precip/Rate: No Precipitation
Precip/Rate: No Precipitation	HN24: HN24 SWE: HN24%:	Temp High:
Temp:	Wind MPH: Select an option Wind Direction: Select an option	Temp Low:
Wind MPH: Select an option	Temp High:	Wind MPH: Select an option
Wind Direction: Select an option	Temp Low:	Wind Direction: Select an option
Peak Gust MPH: Gust Direction: Select an option Gust Time: --:-- --	Peak Gust MPH: Gust Direction: Select an option Gust Time: --:-- --	Peak Gust MPH: Gust Direction: Select an option Gust Time: --:-- --

CLIENT DEMO FEEDBACK

- Auto-populate specific fields
  - Pull previous data from database
  - Auto increment running totals (YTD)
- Focus on the AM Form





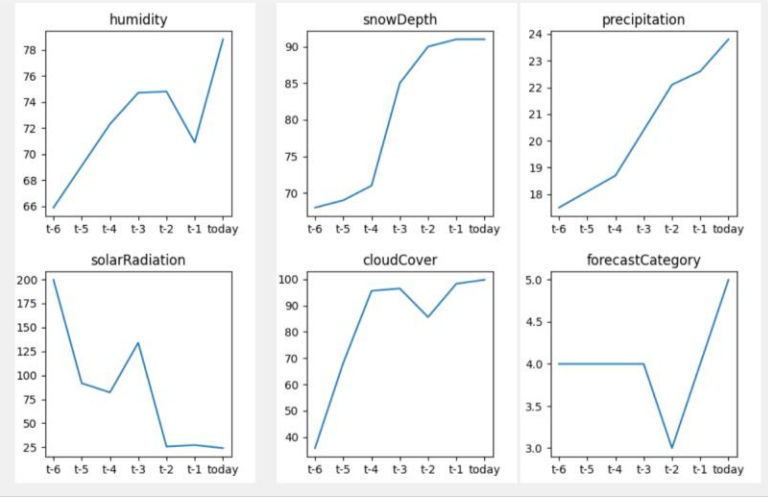
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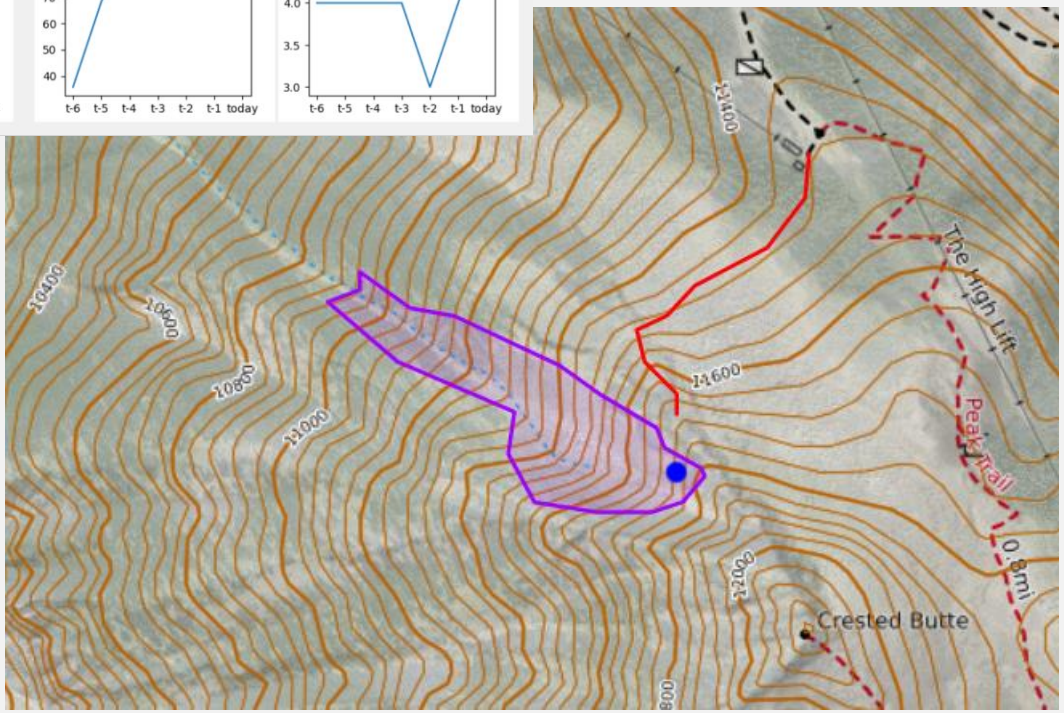
- Client was very happy with initial product
- Forecasters were able to use our product for last 3 weeks of 24-25 season
  - 23 days a usage/data

Date	PM-Form	HS	HN24	HN24 SWE	HST	YTD	YTD SWE	Temp Low	Temp High	Wind Speed	Wind Direction	Peak Gust	DELETE	DELETE
2025-05-04	Update	0.0	None	None	None	None	None	None	None	None	None	None	AM	PM
2025-04-06	Update	68.0	0.0	0.0	None	213.0	18.29	11.0	24.0	light	NW	6.0	AM	PM
2025-04-05	Update	70.0	1.0	0.1	None	213.0	18.29	10.0	22.0	light	NE	5.0	AM	PM
2025-04-04	Update	70.0	2.0	0.1	8.0	212.0	18.19	9.0	18.0	None	None	4.0	AM	PM
2025-04-03	Update	70.0	3.0	0.4	7.0	210.0	18.09	None	None	None	None	None	AM	PM
2025-04-02	Update	67.5	4.0	0.4	6.0	207.0	17.69	4.0	20.0	light	W	9.0	AM	PM
2025-04-01	Update	64.5	2.0	0.15	2.0	203.0	17.29	18.0	30.0	light	SSW	32.0	AM	PM
2025-03-31	Update	61.0	1.0	0.0	0.0	201.0	17.04	16.0	26.0	light	W	0.0	AM	PM
2025-03-30	Update	62.0	0.5	0.1	None	200.0	17.09	19.0	34.0	light	SW	8.0	AM	PM
2025-03-29	Update	64.0	2.0	0.15	2.0	199.5	17.04	23.0	39.0	light	WSW	12.0	AM	PM
2025-03-28	Update	62.5	0.0	0.0	0.0	197.5	16.89	35.0	44.0	light	SW	None	AM	PM
2025-03-27	Update	63.5	0.0	0.0	0.0	197.5	16.89	33.0	47.0	light	SW	None	AM	PM
2025-03-26	Update	66.5	0.0	0.0	0.0	197.5	16.89	29.0	44.0	calm	W	None	AM	PM
2025-03-25	Update	69.0	0.0	0.0	0.0	197.5	16.8	25.0	38.0	light	W	None	AM	PM
2025-03-24	Update	71.0	0.0	0.0	0.0	197.5	16.8	18.0	32.0	light	W	None	AM	PM
2025-03-23	Update	74.0	4	0.35	7.5	197.5	16.89	18.0	24.0	light	W	None	AM	PM
2025-03-21	Update	68.0	4	0.0	0.0	193.5	16.25	5.0	24.0	light	W	None	AM	PM
2025-03-20	Update	69.0	0.0	0.0	0.0	189.5	None	3.0	11.0	calm	SW	None	AM	PM
2025-03-19	Update	70.0	6.0	None	8.0	189.5	None	4.0	22.0	calm	WNW	None	AM	PM
2025-03-18	Update	65.0	2.0	0.15	2.0	183.5	16.0	21.0	33.0	moderate	SSW	None	AM	PM
2025-03-17	Update	66.0	0.0	0.0	0.0	181.5	15.34	7.0	24.0	light	SSW	None	AM	PM
2025-03-16	Update	67.0	3.5	0.3	9.0	181.5	15.42	5.0	15.0	None	None	None	AM	PM
2025-03-15	Update	63.0	3.0	0.2	6.0	178.0	15.12	6.0	15.0	light	WNW	None	AM	PM
2025-03-14	Update	63.0	3.0	1.0	3.0	175.0	15.0	15.0	38.0	light	SW	None	AM	PM

FUTURE WORK

- Dynamic and more in-depth visuals
- A better built PM form
- The ability for them to record more data
- Routes/GPS monitoring
  - Track patrollers while they check/open terrain
  - Avalanche Mitigation tracking
    - Track where shots are placed and results



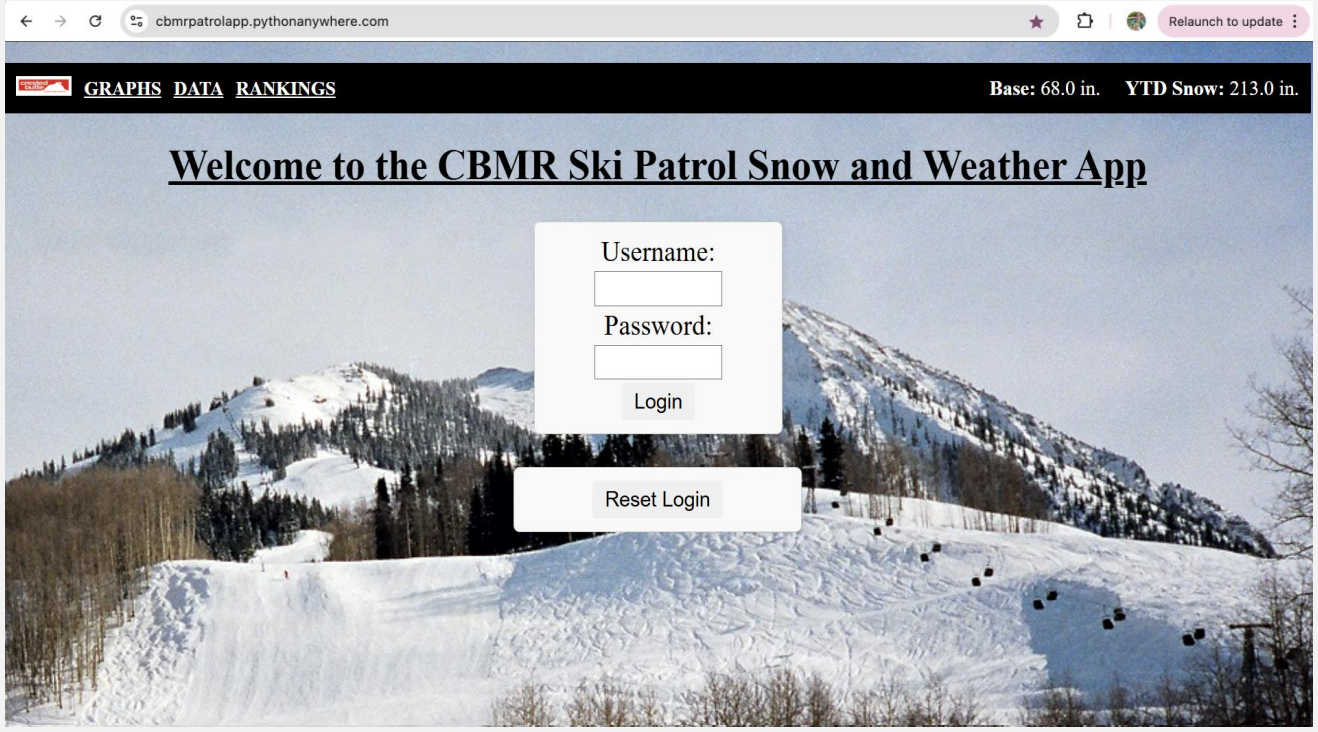


IMPACT

- Our product is working to replace multiple different workflows
  - Microsoft Office, Google Drive, and InfoEx
  - InfoEx costs $1800 per year
- Our product streamlines the AM, PM, and data management workflow
  - Ski patrol spends half the time filling out AM and PM forms
  - Patrollers loved the PDF in the emails they received
- Biggest impact
  - 100% increase in data recall and retention

CONCLUSION

- Created a website for CBMR Ski Patrol
- Implemented an efficient and standardized way to record and retrieve data
- Provided basic visualizations and statistics



THANK YOU

- Crested Butte Ski Patrol
  - Krista Hildebrandt and Scott Sanders
- Western Mathematics and Computer Science
  - Mallory Weber

QUESTIONS

